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The Diffusion Bubble Model for Product Diffusion for Low-Data Product Life Cycle Prediction in Batik Fashion Enterprises: A Comparative Study with Bass Diffusion, Triangle, and Circle Models

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Abstract: Prediction of product life cycle (PLC) at the early stage is important to support the planning of production, inventory control, launch timing, and decisions of product renewal. However, newly introduced batik products often have little historical sales data; therefore, it is difficult to apply conventional forecasting methods. This study introduced the Diffusion Bubble Model for Product Diffusion (DBM-PD) as a low-data diffusion model for predicting cumulative product adoption in batik fashion small and medium enterprises (SMEs). The DBM-PD adapts the bubble-decomposition logic into a product diffusion context, where each adoption bubble represents a possible market response at a specific stage of the PLC. To assess the model performance, this study compared the DBM-PD with three benchmark models: the Bass Diffusion Model, Triangle Model, and Circle Model. Four datasets of PLC from batik fashion products were used as empirical cases. Each dataset was transformed into a cumulative adoption ratio to ensure comparable model evaluation. The results showed that the DBM-PD achieved the lowest prediction error across all four datasets. The Bass Diffusion Model ranked second, indicating that innovation and imitation effects remain relevant in batik product adoption. The Triangle and Circle Models were useful as simple geometric benchmarks, but their assumptions were less able to capture irregular adoption patterns. These findings revealed that the DBM-PD could establish a more flexible prediction structure for estimating PLC at the early stage when only limited data were available. A parameter sensitivity check discovered that the number and width of bubbles affected in-sample accuracy: additional bubbles could reduce errors but also increased model complexity, while the seven-bubble, $\sigma = 0.08$ configuration offered a transparent balance between accuracy and parsimony. The root mean square error (RMSE) rankings were identical across the four datasets, and a Friedman test indicated a significant overall difference among the models ($\chi^2(3) = 12.00$, $p = 0.007$). This study contributes to the management of technology-based PLC by offering a practical framework of diffusion modelling for batik fashion SMEs.

Keywords: Diffusion Bubble Model for Product Diffusion; Product life cycle prediction; Batik fashion small and medium enterprises; Limited data; Bass Diffusion Model; Geometric diffusion; Early-stage; Product diffusion

1 Introduction

Prediction of product life cycle (PLC) is crucial for small and medium enterprises (SMEs) because it supports planning of production, inventory control, product launch decisions, and product renewal strategies. This issue becomes more critical in the fashion industry because fashion products face the challenges of uncertain demands, short PLC, broad product variety, seasonality, and rapid changes in customer preferences. These conditions result in more complex forecasting of fashion demand than many other product categories [1].

Batik fashion products confront similar challenges as discussed. Batik SMEs often introduce new motifs, colors, and product variants to respond to changing market tastes. These new products are usually launched without sufficient historical sales data, leading to a practical forecasting problem, yet many conventional forecasting methods depend on complete and stable past sales records. Early life cycle forecasting is difficult because sales volumes at

the early stage are still small, thus curtailing the accuracy of long-range prediction [2]. Therefore, this limited-data condition propels the need for practical prediction models that could support SMEs during the early stage of product development. For newly launched and short-lived products, firms require operable forecasting approaches provided that there is a lack of historical sales data [2]. Recent research on products with short-life cycles has suggested that limited historical data is a major cause for the difficulties in predicting the sales of newly launched products [3].

Diffusion modelling provides a useful framework for predicting how a new product is adopted over time. The Bass Diffusion Model is one of the most established models for explaining new product diffusion [4], [5]. It represents product adoption through two mechanisms: innovation and imitation. Innovation reflects adoption driven by external influence, while imitation reflects adoption driven by previous adopters and social influence. Prior research used the Bass Diffusion Model for the demand forecasting of a pre-launch new product by combining diffusion modelling with statistical and machine learning approaches [6].

Although the Bass Diffusion Model is widely used, diffusion modelling research continues to develop because the adoption of a product often varies across products, markets, and diffusion stages. A long-term review of innovation diffusion modelling indicated that the field has expanded by modifying existing models and adding greater flexibility to address different forecasting contexts [7]. Moreover, several studies have extended the Bass Diffusion Model to improve its practicality under limited or uncertain market conditions. The quadratic-interval Bass Model, for example, combines interval regression with the Bass diffusion structure to provide sales forecasts with fewer observations than traditional innovation diffusion models [8]. Other studies have improved prediction of diffusion by integrating market-specific variables. A generalized Bass Diffusion Model for electric vehicle diffusion incorporates political, economic, technological, and social variables to explain market-specific adoption drivers while retaining the basic Bass Diffusion Model structure [9]. Recent research has also integrated uncertainty and consumer's sentiment into the Bass Diffusion Model. The uncertain Bass model incorporates uncertainty theory into the conventional Bass diffusion framework to improve the accuracy of new energy vehicle sales forecasting and provide decision support for production planning, risk management, and inventory management [10].

In addition to behavioral diffusion models, this study considered geometric and bubble-based models. The Triangle and Circle Models were employed as simple geometric baselines because they require only basic assumptions of PLC, such as the length and midpoint of a life cycle. These models are suitable for SMEs because they are easy to calculate and do not require complex external variables. They provide a practical comparison against the Bass Diffusion Model in the prediction of early-stage PLC. This study further introduced the Diffusion Bubble Model for Product Diffusion (DBM-PD), referred to be the DBM-PD. The model was adapted from the DBM, which decomposed a signal into multiple isotropic diffusion bubbles to capture hidden diffusion patterns that could not be represented by a single averaged model [11]. In this study, the bubble concept was translated into product diffusion, in which each bubble represented a possible adoption response at a specific stage of the PLC.

The main research gap addressed in this study was the lack of a simple comparative prediction framework for new batik fashion products based on limited data. Existing forecasting approaches often require longitudinal data, large external datasets, social media indicators, or advanced computational resources. These requirements are often difficult to meet, especially batik SMEs. Therefore, this study evaluated whether low-data diffusion models could provide reliable prediction support for early-stage PLC of batik fashion. This study also compared four models: the Bass Diffusion Model, Triangle model, Circle model, and DBM-PD. Four PLC datasets from batik fashion products were used as empirical cases. Each dataset was transformed into a cumulative adoption ratio to allow fair comparison across models. Model performance was evaluated using prediction error metrics, with root mean square error (RMSE) being the main accuracy measure.

This study contributes to the management of technology-based PLC in three ways. First, it provided a comparative evaluation of behavioral, geometric, and bubble-based diffusion models for the prediction of batik fashion products. Second, it developed the DBM-PD as an adapted low-data diffusion model for the analysis of the PLC. Third, it offered a practical prediction framework that could support SMEs in production planning, stock control, product launch timing, and new product renewal decisions. The novelty of this study lied in the development of a low-data product diffusion framework for the prediction of batik fashion PLC at its early stage. Unlike conventional forecasting approaches that depend on long historical sales records, the proposed framework evaluated simple diffusion models that could be adapted for SMEs with the availability of limited data. The framework was deemed useful for planning new products in batik fashion SMEs.

The remainder of this paper is organized as follows. Section 2 reviews the prediction of PLC, diffusion models, and low-data forecasting. Section 3 explains the research methodology, including data normalization, parameter setting, sensitivity analysis, and model evaluation. Section 4 presents and discusses the empirical results, statistical comparison, and examples of practical implementation. Section 5 concludes the study and provides future research directions.

2 Literature Review

2.1 Prediction of the Product Life Cycle of Fashion Products

PLC prediction has become an important topic in fashion research because it affects production planning, inventory control, product launch timing, and product renewal decisions. Fashion products are difficult to predict because their demand is shaped by short product cycles, high product variety, seasonal effects, and fast changes in consumer preferences [1]. Most studies in this area tried to improve forecasting accuracy by using statistical models, artificial intelligence, or machine learning. These approaches are useful in data-rich firms, but they are not always suitable for SMEs that do not have complete and structured sales records.

A key issue in PLC prediction is the lack of early sales data. Many forecasting models need enough historical observations to identify patterns of demand. This creates a problem for new products because their market response is still uncertain at the early stage. Li et al. [2] showed that limited early sales data could undermine the reliability of forecasts for long-range PLC. Although their study focused on consumer technology products, the same problem could occur in fashion products, especially when new designs were launched frequently.

Recent studies on short-life-cycle products have attempted to solve this problem through machine learning-based forecasting frameworks [3]. These models could improve prediction when product attributes, comparable products, and historical transaction data were available. However, their application in SMEs remains limited. Many SMEs do not have the data infrastructure, technical capacity, or computational resources to apply these methods, hence generating a gap between advanced forecasting methods and the practical needs of small firms.

This gap is obvious in batik fashion SMEs. Batik products often change in motif, color, design, and market appeal. New variants may be introduced before the firm has enough sales history to estimate demand using conventional forecasting methods. Therefore, the main issue is not only how to improve the accuracy of prediction, but also how to develop a model that is simple enough to use with limited data. This study addressed this issue by evaluating low-data diffusion models that require simple parameters and can support prediction of PLC at its early stage for new batik fashion products.

2.2 Diffusion Modelling and the Bass Diffusion Model

Diffusion modelling has long been employed to explain how new products enter and spread within a market. The basic assumption is that adoption does not occur simultaneously for all potential buyers. Instead, it develops gradually through external influence and interaction among users. The Bass Diffusion Model is one of the most influential models in this area because it offers a simple way to represent these two mechanisms. In this model, innovators adopt a product independently from previous users, while imitators are influenced by the number of people who have already adopted the product [12]. This interrelationship renders the model useful for describing cumulative adoption and estimating patterns of PLC.

Despite its widespread use, the Bass Diffusion Model has its limitations. Its standard form assumes that the coefficients of innovation and imitation remain constant during the diffusion process. This assumption may not always fit real market conditions, especially industries where customer preferences, intensity of promotion, product visibility, and social influence are volatile. In fashion markets, for example, demand could shift because of trend exposure, seasonality, design variation, or short-term customer attention. Therefore, the Bass Diffusion Model is useful as a theoretical foundation, but its direct application requires careful evaluation when the product has a short and unstable life cycle.

The evolution of diffusion modelling shows that researchers have continued to adjust the Bass Diffusion Model to fit different empirical contexts. A review of innovation diffusion models revealed that the main diffusion models had already been established by the 1970s, but later studies expanded them by adding new parameters, flexible assumptions, and improved forecasting procedures [7]. This indicates that the value of the Bass Diffusion Model lies not only in its original equation, but also in its ability to serve as a base model for further development. However, these extensions often require additional data, such as price, advertising, product attributes, or market-level variables.

Pre-launch forecasting is one relevant extension for novel product research. Lee et al. [6] combined the Bass Diffusion Model with statistical and machine learning approaches to estimate new product demand prior to the availability of complete sales data. Their study showed that diffusion parameters could be estimated earlier by using product attributes and diffusion patterns from comparable products. This approach was important because it resolved the problem of incomplete sales history. However, it may still be difficult for SMEs because it requires supporting data from comparable products and a more advanced modelling process.

Based on this literature, the Bass Diffusion Model remains a strong reference for product diffusion analysis, but it is not always sufficient for low-data forecasting in SMEs. The model offers clear behavioral interpretation through innovation and imitation effects, yet its estimation of parameters could become challenging when only limited early-stage data are available. This study therefore used the Bass Diffusion Model as a behavioral diffusion benchmark and compared it with simpler geometric models and the DBM-PD model. This comparison was essential to assess whether a low-data prediction of PLC framework could provide practical support for new batik fashion products.

2.3 Extension of the Bass Diffusion Model under Uncertainty and Market Signals

The standard Bass Diffusion Model provides a clear structure for explaining product adoption, but its deterministic form has been widely criticized for being too restrictive in dynamic market settings. Product diffusion rarely depends only on a fixed innovation coefficient and a fixed imitation coefficient. In practice, adoption could change because of word-of-mouth, market uncertainty, customer sentiment, digital exposure, price changes, and external information flows. Peres et al. [13] argued that diffusion research had moved beyond the original deterministic model because contemporary markets involved stronger social interaction, network effects, and heterogeneous consumer responses. This indicates that the Bass Diffusion Model is useful as a foundation, but its assumptions need adaptation when market signals are unstable.

One stream of research extended the Bass Diffusion Model by addressing uncertainty and scarcity of data. The quadratic-interval Bass model combines interval regression with the Bass diffusion structure to estimate new product sales under little observations [14]. This approach is important because it recognizes that early diffusion data may not be precise enough for conventional point estimation. Instead of forcing one fixed parameter value, interval-based modelling allows a range of possible diffusion outcomes. However, this method only focuses on improving the Bass structure itself. It does not prove whether simpler non-Bass models may be more practical when SMEs face limited data and restricted modelling capability.

A second stream incorporated consumer sentiment and market noise into diffusion modelling. The uncertain Bass model introduces uncertainty theory into the conventional Bass diffusion framework to capture market disturbances and improve new energy vehicle sales forecasting [10]. This approach is valuable because it treats imitation as a time-sensitive market response rather than a constant parameter. It also provides scenario-based output that could support managerial decisions. However, the model requires online review data, sentiment extraction, and uncertainty modelling. These requirements may confine its direct application for batik SMEs that do not have large digital review datasets or technical capacity for sentiment analysis.

A third stream used social media indicators to enrich Bass-based diffusion analysis. Ding et al. [15] integrated sentiment values, Recency, Frequency, and Monetary (RFM) indicators, long-term attention, and an improved Bass model to evaluate communication diffusion when financial indicators were difficult to access. Their study demonstrated that diffusion modelling could still be useful when direct sales or revenue data were incomplete. However, the context was media communication rather than adoption of physical products. The model depended on comment data from social media, which may not be available or consistent for small batik producers.

Overall, these extensions implied that Bass-based models could be improved by adding uncertainty, sentiment, social influence, and external market signals. Yet, they pointed towards a practical gap. Many improved models become more data-intensive and technically demanding as they become more realistic. This creates a mismatch between advanced diffusion research and the operational needs of SMEs. For this reason, the present study did not only extend the Bass Diffusion Model but also compared the former model with the Triangle, Circle, and DBM-PD models to evaluate which approach could support PLC prediction with limited data in the early stage.

2.4 Geometric Diffusion Models for Low-Data Prediction

Geometric models offer a different perspective from behavioral diffusion models. While the Bass Diffusion Model explains adoption through innovation and imitation, geometric models focus on the expected shape of cumulative adoption. Their main value lies in simplicity. They could be applied when only the estimated length of PLC, its midpoint, and target of cumulative adoption are available. These enable them to become relevant for low-data prediction in SME settings, where detailed market variables and long historical sales records are often missing.

The Triangle model represents cumulative adoption as a simple symmetric process. It assumes that adoption grows gradually before the midpoint of the life cycle and then moves toward saturation after the midpoint. This structure is effective as a transparent baseline but its main constraint is the assumption of symmetry. In real fashion markets, adoption may rise earlier, peak later, or shift because of promotion, seasonality, reseller activity, or customer trend response. Kendall [16] provided a mathematical foundation for triangle-based diffusion by illustrating that Bookstein triangle shape could be represented within a stochastic dynamical framework. Although the scholar did not focus on product adoption, triangle geometry was used as a structured form for diffusion-like modelling.

The Circle Model is found to be a smoother geometric alternative. It represents cumulative adoption through the proportional area of a circular segment, which allows adoption to move gradually from the early stage toward market saturation. However, the Circle Model does not explain why customers adopt the product. It only describes the shape of cumulative adoption. According to Majumdar and Laha [17], circular diffusion represented bounded dynamic processes through Brownian motion on the circle and the von Mises diffusion process. Their work focused on stochastic correlation, but it supported the broader methodological idea that circular structures could model dynamic processes within a bounded domain.

In this study, the Triangle and Circle Models were employed as low-data geometric benchmarks but not as complete behavioral theories of market adoption. They helped evaluate whether adoption of batik products could

be approximated by simple life cycle shapes. If these models produced low prediction errors, the adoption process might follow a regular and predictable pattern. If they performed less satisfactory than the Bass Diffusion Model or DBM-PD, the results suggested that adoption of batik products required a more flexible diffusion structure. This comparison was important because it distinguished simple geometric regularity from more complex adoption behavior under limited-data conditions.

2.5 Bubble-Based Diffusion Modelling and Diffusion Bubble Model for Product Diffusion

Decomposition-based modelling offers a useful perspective when a diffusion pattern could not be well represented by a single curve. In this approach, an observed process is treated as the result of several underlying components rather than one homogeneous mechanism. This treatment is particularly relevant for prediction of PLC because market adoption may occur in several ways. A product could be first adopted by loyal customers, then by broader consumers, and later by buyers affected by promotion, reseller activity, or seasonal demand.

The DBM was originally introduced in medical imaging as a spectrum-based framework. Zhang et al. [11] developed the DBM to decompose a diffusion MRI signal into multiple isotropic "bubbles" and optional anisotropic components. The purpose was to capture hidden diffusion heterogeneity that could be missed by a single averaged model. The original DBM was therefore not a product forecasting model. Its contribution lied in the decomposition logic, where a complex diffusion signal was represented as a weighted combination of simpler diffusion elements.

This study adapted that logic in product diffusion through the DBM-PD. In the DBM-PD, each bubble represents a possible adoption response located at a specific stage of the PLC. The predicted cumulative adoption curve is formed by combining these bubbles through estimated weights. This adaptation allows the model to represent staged adoption behavior without assuming that the whole PLC follows one fixed S-curve, triangle curve, or circular curve.

The DBM-PD should be positioned carefully because it is neither a direct transfer of the medical DBM model nor a behavioral substitute for the Bass Diffusion Model. The Bass Diffusion Model is an aggregate behavioral diffusion model: one cumulative S-curve is generated by the innovation coefficient p, the imitation coefficient q, and market potential, so its parameters have a direct market-adoption interpretation. The DBM-PD is a representation of the decomposition-based curve. It reconstructs cumulative adoption as a convex combination of several localized diffusion components, and the estimated bubble weights indicate the relative contribution of early-, middle-, and late-stage adoption responses rather than innovation or imitation propensities. Consequently, Bass provides stronger behavioral interpretability, whereas the DBM-PD offers greater structural flexibility for staged or irregular adoption. The comparison in this study further examined whether the additional flexibility could improve low-data fitting without claiming that the DBM-PD replaced the theoretical mechanisms of the Bass Diffusion Model.

2.6 Research Gap

The literature featured four important points as shown in Table 1. Fashion demand forecasting is difficult owing to its short dynamic product life cycle and insufficient early sales data. Although the Bass model dominates new product diffusion modeling, various extensions have been proposed to tackle practical uncertainties and data limitations. For data-scarce SMEs, geometric and bubble-based models offer lightweight and feasible forecasting alternatives.

Table 1. Comparative analysis of related works and the proposed research

Reference	F/SLC	LD	ES/NPP	Bass	Bass-Ext/MS	Geo. Model	Bub. Model	SME	Model Comp.	Context
[1]	✓									Fashion products
[2]		✓	✓							Consumer technology products
[3]		✓	✓							Short-life-cycle new products
[12]				✓						Consumer durables
[7]				✓	✓					Innovation diffusion models
[6]		✓	✓	✓					✓	New product demand forecasting
[13]				✓	✓					New product growth models
[14]		✓	✓	✓	✓					New product sales diffusion
[10]				✓	✓				✓	New energy vehicles
[15]		✓		✓	✓				✓	TV content communication diffusion
[16]						✓				Triangle shape diffusion
[17]						✓				Circular diffusion/stochastic correlation
[18]				✓	✓		✓			PLC quantification
[19]	✓	✓	✓						✓	Fast fashion
[20]	✓									Fashion sales forecasting
[21]		✓	✓						✓	New product demand profiles
[22]				✓	✓				✓	Product sales forecasting with online reviews
This study	✓	✓	✓	✓		✓	✓	✓	✓	**New batik fashion products**

Note: F/SLC = Fashion/Short-life Cycle; LD = Limited Data; ES/NPP = Early-Stage/New Product Prediction; Bass-Ext/MS = Bass Extension/Market Signals; Geo. Model = Geometric Model; Bub. Model = Bubble-Based Model; Model Comp. = Comparison of the Models; SMEs = Small and medium enterprises; PLC = Product life cycle.

This paper positioned itself as a technology-based low-data PLC prediction study, not only tested model accuracy but also compared assumptions of different diffusion. The Bass Diffusion Model represents behavioral diffusion through innovation and imitation whereas the Triangle and Circle Models represent simple geometric diffusion. The proposed DBM-PD stands for flexible bubble-based diffusion. A comparative study helped identify the optimal model to support early-stage product planning for batik fashion SMEs.

3 Methodology

This study applied a quantitative modelling approach to compare four diffusion models in PLC prediction at the early stage for batik fashion products as shown in Figure 1. The models evaluated in this study were the Bass Diffusion Model, Triangle Model, Circle Model, and DBM-PD. The purpose of the comparison was not only to identify the most accurate model but also to evaluate whether low-data diffusion models could support PLC prediction when complete historical sales data were not available.

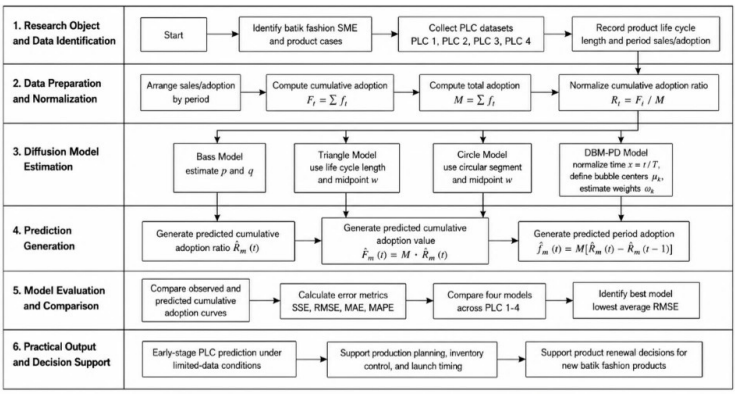


Figure 1. Proposed methodology for low-data PLC prediction in batik fashion SMEs

Note: DBM-PD = Diffusion Bubble Model for Product Diffusion; SMEs = Small and medium enterprises; PLC = Product life cycle; SSE = Sum of squared errors; RMSE = Root mean squared error; MAE = Mean absolute error; MAPE = Mean absolute percentage error.

The study utilized four PLC datasets from batik fashion products produced by a fashion-based SME. Each dataset represented one case of product life cycle. The four datasets were treated as empirical product-level cases that allowed comparison across different patterns of adoption. This methodological design was appropriate because the research could focus on model comparison and practical support for prediction, not population-level generalization.

The methodological process consisted of five stages. First, data on PLC were organized by period. Second, actual sales or adoption data were transformed into cumulative adoption ratios. Third, four diffusion models were applied to estimate the ratio of predicted cumulative adoption. Fourth, model errors were calculated by comparing observed and predicted cumulative adoption ratios. Fifth, model performance was evaluated using error metrics.

3.1 Data Analysis

The unit of analysis was the PLC of an individual batik fashion product. The four cases involved distinct product variants differentiated by the SMEs through design, motif, color, or related combinations of style. To protect commercial confidentiality, the products were reported as PLC 1–PLC 4; brand identifiers, motif codes, prices, and absolute sales volumes were not disclosed. The SMEs supplied aggregated period-level sales/adoption records from product launch till the end of active selling, while the analysis did not use customers' personal data.

The cases were selected by four criteria: (1) a clearly identifiable launch period; (2) consecutive records without any missing intervals; (3) positive total adoption; and (4) a sufficiently complete trajectory reaching near-saturation or formal product withdrawal. It was found that four products had met all criteria. Since the original SME database used a consistent operational sales interval but this commercial interval itself was confidential, the study reported the time index generically as "period" rather than imposing a weekly or monthly label. The cases were therefore considered to be appropriate for model comparison only but were not intended to represent the full population of batik fashion products.

The length and midpoint of PLC used in the geometric models are shown in Table 2.

The midpoint value was used as the main parameter in the Triangle and Circle Models. In the DBM-PD model, the period was normalized into a 0 to 1 scale to allow bubble centers to be placed at comparable stages across all PLC. The same normalized time scale and parameter configuration were applied to all four products so that differences in model performance were not caused by product-specific tuning of bubble positions or widths.

Table 2. Length and midpoint of product life cycle (PLC)

Dataset	Length of Product Life Cycle	Midpoint Value
PLC 1	32 periods	16
PLC 2	32 periods	16
PLC 3	28 periods	14
PLC 4	20 periods	10

3.2 Data Transformation

The original sales data were transformed into ratios of cumulative adoption. This transformation was required because each product had a different total sales volume and length of PLC. Ratio-based transformation allowed all models to be compared on the same scale. For product j at period t, actual sales or adoption is denoted as $f_{j,t}$.

The cumulative adoption is calculated as:

$$F_{j,t} = \sum_{i=1}^{t} f_{j,i}$$

The total adoption during the observed PLC is calculated as:

$$M_j = \sum_{t=1}^{T_j} f_{j,t}$$

The cumulative adoption ratio is calculated as:

$$R_{j,t} = \frac{F_{j,t}}{M_j}$$

where, T_j is the length of PLC for product j, $F_{j,t}$ is the cumulative adoption at period t, M_j is the total adoption, and $R_{j,t}$ is the observed ratio of cumulative adoption. The value of $R_{j,t}$ ranges from 0 to 1.

The predicted cumulative adoption ratio generated by each model is denoted as $\widehat{R}_{m,j,t}$, where m represents the prediction model.

3.3 Bass Diffusion Model

The Bass Diffusion Model suggests product adoption through two behavioral mechanisms: innovation and imitation. Innovation refers to adoption caused by external influence, whereas imitation means adoption caused by previous adopters or social interaction. In this study, the Bass Diffusion Model, being the behavioral diffusion benchmark, had its cumulative adoption ratio expressed as:

$$\widehat{R}_{B,j}(t) = \frac{1 - e^{-(p_j+q_j)t}}{1 + \frac{q_j}{p_j} e^{-(p_j+q_j)t}}$$

where, p_j is the innovation coefficient, q_j is the imitation coefficient, and t is the period of PLC. The predicted value of cumulative adoption was calculated as:

$$\widehat{F}_{B,j}(t) = M_j \hat{R}_{B,j}(t)$$

The predicted adoption in each period was calculated as:

$$\hat{f}_{B,j}(t) = M_j \left[\hat{R}_{B,j}(t) - \hat{R}_{B,j}(t-1) \right]$$

The parameters p_j and q_j are estimated by minimizing the squared error between the observed cumulative adoption ratio and the predicted cumulative adoption ratio:

$$\min_{p_j, q_j} \sum_{t=1}^{T_j} \left(R_{j,t} - \hat{R}_{B,j}(t) \right)^2$$

subject to:

$$p_j > 0, q_j > 0$$

3.4 Triangle Model

The Triangle Model was used as a simple geometric benchmark. It assumes that cumulative adoption follows a symmetric pattern of life cycle. Adoption increases during the first half of the PLC and moves toward saturation during the second half. This model uses the midpoint of the PLC as its main parameter. Let w_j be the midpoint of the PLC:

$$w_j = \frac{T_j}{2}$$

The Triangle Model was expressed as:

$$\hat{R}_{\Delta,j}(t) = \begin{cases} \frac{1}{2}\left(\frac{t}{w_j}\right)^2, & 0 \le t \le w_j \\ 1 - \frac{t}{w_j} + \frac{t^2}{2w_j^2}, & w_j < t \le 2w_j \end{cases}$$

where, $\widehat{R}_{\Delta,j}(t)$ is the predicted cumulative adoption ratio of the Triangle Model. The predicted cumulative adoption value was calculated as:

$$\widehat{F}_{\Delta,j}(t) = M_j \hat{R}_{\Delta,j}(t)$$

The predicted adoption in each period was calculated as:

$$\hat{f}_{\Delta,j}(t) = M_j \left[\hat{R}_{\Delta,j}(t) - \hat{R}_{\Delta,j}(t-1)\right]$$

The Triangle Model does not estimate behavioral parameters. Its main inputs are the length of PLC and midpoint. This renders the model suitable as a low-data benchmark.

3.5 Circle Model

The Circle Model was also used as a geometric benchmark. It represents cumulative adoption through the proportional area of a circular segment. The model assumes that adoption follows a smoother cumulative pattern than the Triangle Model.

The midpoint of the product life cycle was used as the circle radius:

$$w_j = \frac{T_j}{2}$$

The circular segment area at period t was calculated as:

$$A_{C,j}(t) = w_j^2 \cos^{-1}\left(\frac{w_j - t}{w_j}\right) - (w_j - t)\sqrt{2w_j t - t^2}$$

The predicted cumulative adoption ratio of the Circle Model is:

$$\hat{R}_{C,j}(t) = \frac{w_j^2 \cos^{-1}\left(\frac{w_j - t}{w_j}\right) - (w_j - t)\sqrt{2w_j t - t^2}}{\pi w_j^2}$$

where, $\widehat{R}_{C,j}(t)$ is the predicted cumulative adoption ratio of the Circle Model. The predicted cumulative adoption value is:

$$\widehat{F}_{C,j}(t) = M_j \widehat{R}_{C,j}(t)$$

The predicted adoption in each period is:

$$\hat{f}_{C,j}(t) = M_j \left[\hat{R}_{C,j}(t) - \hat{R}_{C,j}(t-1)\right]$$

The Circle Model was included to evaluate whether batik product adoption followed a smooth geometric accumulation pattern.

3.6 Diffusion Bubble Model for Product Diffusion

The DBM-PD was adapted from the bubble-decomposition logic of the DBM. The original DBM was developed as a spectrum-based model that decomposed a signal into multiple isotropic diffusion bubbles and optional anisotropic components to capture hidden heterogeneity that might not be represented by a single averaged model. Instead, it adapted the bubble-spectrum logic to product diffusion, where each bubble represents a possible adoption response located at a specific stage of the product life cycle and the cumulative curve is estimated as a weighted combination of these responses.

The normalized PLC was defined as:

$$x_{j,t} = \frac{t}{T_j}$$

where, $x_{j,t}$ ranges from 0 to 1. A set of K bubble centers was defined as:

$$\mu_k \in \{0.05, 0.20, 0.35, 0.50, 0.65, 0.80, 0.95\}$$

The standard deviation of each bubble was fixed as:

$$\sigma = 0.08$$

The unnormalized bubble response was defined as:

$$G_k\left(x_{j,t}\right) = \exp\left[\left(-\frac{1}{2}\left(\frac{x_{j,t} - \mu_k}{\sigma}\right)^2\right)\right]$$

Each bubble response was normalized, so that it formed a period-share pattern:

$$B_k\left(x_{j,t}\right) = \frac{G_k\left(x_{j,t}\right)}{\sum_{t=1}^{T_j} G_k\left(x_{j,t}\right)}$$

The cumulative bubble response is:

$$C_k\left(x_{j,t}\right) = \sum_{\tau=1}^{t} B_k\left(x_{j,\tau}\right)$$

The predicted cumulative adoption ratio of DBM-PD was calculated as:

$$\hat{R}_{D,j}(t) = \sum_{k=1}^{K} \omega_{j,k} C_k\left(x_{j,t}\right)$$

where, $\omega_{j,k}$ is the weight of bubble k for product j. The weights were estimated using constrained non-negative least squares:

$$\min_{\omega_{j,k}} \sum_{t=1}^{T_j} \left(R_{j,t} - \hat{R}_{D,j}(t)\right)^2$$

subject to:

$$\omega_{j,k} \geq 0$$

$$\sum_{k=1}^{K} \omega_{j,k} = 1$$

The baseline configuration used $K = 7$ bubbles with fixed centers $\mu = \{0.05, 0.20, 0.35, 0.50, 0.65, 0.80, 0.95\}$ and a common standard deviation $\sigma = 0.08$. The centers were specified a priori at approximately equal intervals to cover the complete normalized life cycle from early launch to late saturation. Fixing the centers avoided productspecific location optimization and rendered the four cases directly comparable. The selected width created overlap between adjacent bubbles, thus permitting a smooth cumulative curve while retaining separation between the stages of adoption.

Only the non-negative bubble weights were estimated separately for each PLC dataset. The estimation minimized squared error subject to ω_j, $k \geq 0$ and $\sum k\omega_j$, $k = 1$. Uniform weights were used as the numerical starting point, and optimization continued until the change in the objective function was below 10^{-8}. Since the objective was quadratic in the weights and the constraints defined a convex simplex, the fitted solution was stable with respect to the starting weights. The unit-sum constraint guaranteed that the predicted cumulative adoption ratio converged to 1 at the end of the observed life cycle.

The predicted cumulative adoption value is:

$$\widehat{F}_{D,j}(t) = M_j \widehat{R}_{D,j}(t)$$

The predicted adoption in each period is:

$$\hat{f}_{D,j}(t) = M_j \left[\hat{R}_{D,j}(t) - \hat{R}_{D,j}(t-1)\right]$$

The DBM-PD was expected to capture staged adoption patterns because different bubbles could represent different phases of market response, such as early adoption from loyal customers, broader market adoption, and late-stage demand renewal.

3.7 Model Evaluation

Model performance was evaluated by comparing the observed ratio of cumulative adoption with the predicted ratio of cumulative adoption. The period error was defined as:

$$e_{m,j,t} = R_{j,t} - \hat{R}_{m,j,t}$$

The sum of squared error (SSE) is:

$$\mathrm{SSE}_{m,j} = \sum_{t=1}^{T_j} \left(R_{j,t} - \hat{R}_{m,j,t} \right)^2$$

The RMSE is:

$$\mathrm{RMSE}_{m,j} = \sqrt{\frac{1}{T_j} \sum_{t=1}^{T_j} \left(R_{j,t} - \hat{R}_{m,j,t} \right)^2}$$

The mean absolute error (MAE) is:

$$\mathrm{MAE}_{m,j} = \frac{1}{T_j} \sum_{t=1}^{T_j} \left| R_{j,t} - \hat{R}_{m,j,t} \right|$$

The mean absolute percentage error (MAPE) is:

$$\mathrm{MAPE}_{m,j} = \frac{100}{T_j} \sum_{t=1}^{T_j} \left| \frac{R_{j,t} - \hat{R}_{m,j,t}}{R_{j,t}} \right|$$

RMSE was used as the main evaluation metric because it gave greater penalty to larger prediction errors. MAE was used to measure average absolute deviation, while MAPE was used to provide percentage-based interpretation. If $R_{j,t} = 0$, the period was excluded from the MAPE calculation to avoid division by zero.

3.8 Parameter Sensitivity and Statistical Comparison

A brief sensitivity analysis was conducted to evaluate whether the DBM-PD performance depended strongly on the number and width of bubbles. The number of bubbles varied over $K \in \{5, 7, 9\}$, and the common standard deviation varied over $\sigma \in \{0.06, 0.08, 0.10\}$. For each K, bubble centers were evenly distributed between 0.05 and 0.95, and the weights were re-estimated using the same non-negative unit-sum constraints. The average RMSE across the four PLC datasets was used to compare parameter settings. Lower RMSE indicates better in-sample fit, while larger K represents greater model complexity and a higher potential for overfitting.

To evaluate whether the observed performance ranking was consistent across datasets, the four RMSE values for each model were treated as paired observations. A Friedman test was adopted because the number of datasets was small and normality could not be assumed. Kendall's W was reported as the rank-consistency effect size. Exact Wilcoxon signed-rank comparisons between DBM-PD and each benchmark were also examined, but their p-values were interpreted cautiously because only four paired cases were available.

3.9 Procedures of Model Comparison

The model comparison was conducted for each PLC dataset. For each such dataset, the Bass Diffusion, Triangle, Circle, and DBM-PD Models generated predicted cumulative adoption ratios. The prediction errors were then calculated using SSE, RMSE, MAE, and MAPE. The best model for each dataset was selected based on the lowest RMSE. In addition to descriptive error metrics, the cross-dataset RMSE rankings were evaluated using the Friedman test and Kendall's W, followed by exact pairwise Wilcoxon signed-rank tests as a small-sample diagnostic.

The average model performance across all PLC datasets was calculated as:

$$\bar{\mathrm{RMSE}}_m = \frac{1}{J} \sum_{j=1}^{J} \mathrm{RMSE}_{m,j}$$

where, $J = 4$ is the number of PLC datasets. The model with the lowest average RMSE was considered the best overall prediction model.

4 Results and Discussion

4.1 Results of Model Fitting

This study evaluated four diffusion models for predicting the pattern of cumulative PLC of batik fashion products. The models were tested on four PLC datasets, namely PLC 1, PLC 2, PLC 3, and PLC 4. Each dataset was transformed into a cumulative adoption ratio before model estimation. This transformation allowed all models to be compared on the same scale, regardless of differences in total sales volume and the length of PLC.

The four models produced different levels of prediction accuracy. Table 3 presents the RMSE values for each model across the four PLC datasets.

Table 3. Comparison of RMSE values across the four prediction models

Dataset	Bass Diffusion Model	Triangle Model	Circle Model	DBM-PD	Best Model
PLC 1	0.0251	0.1383	0.0869	0.0061	DBM-PD
PLC 2	0.0226	0.2263	0.1431	0.0070	DBM-PD
PLC 3	0.0234	0.0993	0.0499	0.0090	DBM-PD
PLC 4	0.0304	0.1899	0.1055	0.0128	DBM-PD

Note: RMSE = Root mean squared error; DBM-PD = Diffusion Bubble Model for Product Diffusion; PLC = Product life cycle.

The results revealed that the DBM-PD achieved the lowest RMSE in all four datasets. The model produced RMSE values of 0.0061 for PLC 1, 0.0070 for PLC 2, 0.0090 for PLC 3, and 0.0128 for PLC 4. These values were consistently lower than those of the Bass Diffusion, Triangle, and Circle models. This indicated that the DBM-PD provided the closest fit to the observed cumulative adoption curves.

Among the three original models, the Bass Diffusion Model produced the best performance. Its RMSE values ranged from 0.0226 to 0.0304 across the four datasets. This proved that the Bass Diffusion Model was more accurate than the Triangle and Circle Models in capturing the patterns of batik product adoption. The result suggested that product adoption in batik fashion products was aptly represented by behavioral diffusion than by simple geometric assumptions.

4.2 Average Model Performance

The average performance of each model is shown in Table 4. The comparison adopted RMSE, MAE, and MAPE to provide a broader view of prediction accuracy.

Table 4. Average prediction error across the four PLC datasets

Model	Average RMSE	Average MAE	Average MAPE (%)
Bass Diffusion Model	0.0254	0.0215	24.44
Triangle Model	0.1635	0.1338	29.21
Circle Model	0.0964	0.0813	55.39
DBM-PD	0.0087	0.0064	10.75

Note: RMSE = Root mean squared error; DBM-PD = Diffusion Bubble Model for Product Diffusion; PLC = Product life cycle; MAE = Mean absolute error; MAPE = Mean absolute percentage error.

The DBM-PD achieved the lowest average RMSE, MAE, and MAPE. Its average RMSE was 0.0087, which was lower than the Bass Diffusion Model at 0.0254, the Circle model at 0.0964, and the Triangle model at 0.1635. This result confirmed that the DBM-PD produced the strongest fitting performance across the four PLC datasets.

The Bass Diffusion Model ranked second based on average RMSE and MAE. This was an important result because the Bass Diffusion Model used only two behavioral parameters, namely the innovation coefficient and imitation coefficient. Its relatively low error showed that the adoption pattern of batik fashion products contained both innovation and imitation effects. In practical terms, this means that adoption may be influenced by early product appeal, customer interaction, word-of-mouth, and market response.

The Circle Model performed better than the Triangle Model in terms of average RMSE and MAE. This indicated that a smoother geometric structure was more suitable than a simple symmetric triangle structure for representing batik product adoption. However, the Circle Model produced higher errors than the Bass Diffusion Model and the DBM-PD. This suggests that a smooth geometric curve alone is not sufficient to capture the observed adoption pattern.

The Triangle Model produced the highest average RMSE (Figure 2). This result indicated that the assumption of symmetric life cycle was too restrictive for batik fashion products. In real market conditions, product adoption

may not rise and decline in a balanced way. Adoption may increase faster or slower depending on product design, customers' preference, seasonal demand, reseller activity, and market exposure.

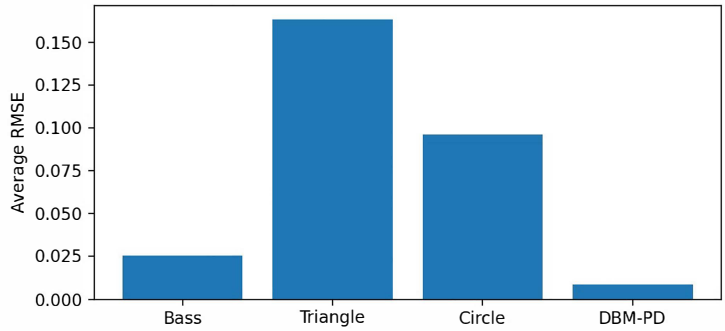


Figure 2. Comparison of average RMSE across the four PLC datasets

Note: RMSE = Root mean squared error; PLC = Product life cycle; DBM-PD = Diffusion Bubble Model for Product Diffusion.

4.3 Parameter Sensitivity and Statistical Comparison

Table 5 summarizes the analysis of the DBM-PD sensitivity analysis. The results confirmed that choices of the parameters influenced fitting performance. Five bubbles produced substantially larger errors, indicating insufficient flexibility for several staged adoption patterns. Increasing the model to nine bubbles reduced in-sample RMSE for narrow bubbles, but it also introduced two additional estimated weights and therefore greater overfitting risk, particularly for PLC 4 with only 20 observations.

Table 5. Sensitivity of the DBM-PD to the number and width of bubbles

Number of Bubbles (K)	Standard Deviation (σ)	Average RMSE	Interpretation
5	0.06	0.0224	Too few and too narrow; underfitting
5	0.08	0.0172	Lower complexity but higher error
5	0.10	0.0177	Smoother but still underfitting
7	0.06	0.0098	Sharp local responses
7	0.08	0.0087	Baseline balance of fit and parsimony
7	0.10	0.0102	Moderate over-smoothing
9	0.06	0.0068	Lowest in-sample error; highest complexity
9	0.08	0.0083	Comparable fit with more parameters
9	0.10	0.0103	Additional bubbles offset by smoothing

Note: DBM-PD = Diffusion Bubble Model for Product Diffusion; RMSE = Root mean squared error.

The $K = 7$ and $\sigma = 0.08$ specification was retained for the main comparison because it remains substantially more accurate than the benchmark models, while using fewer weights than the nine-bubble alternatives. The sensitivity results therefore supported the robustness of the main conclusion, but they also showed that the DBM-PD accuracy should always be reported together with its parameter configuration and complexity.

The RMSE ranking was identical for all four datasets: the DBM-PD ranked first, Bass Diffusion second, Circle third, and Triangle fourth. The Friedman test confirmed an overall difference among the four models, $\chi^2(3) = 12.00$, $p = 0.007$, with Kendall's $W = 1.00$, indicating complete rank agreement across the cases. Relative to the average RMSE of the Bass Diffusion, Circle, and Triangle Models, the DBM-PD reduced RMSE by 65.7%, 91%, and 94.7%, respectively. Exact pairwise Wilcoxon tests produced $p = 0.125$ for each DBM-PD comparison because four datasets were insufficient for an individually significant two-sided exact test even when every difference favored the DBM-PD. Thus, the overall rank test and effect magnitudes supported a consistent performance advantage, while the pairwise inference remained limited by the sample size.

4.4 Observed and Predicted Cumulative Adoption Curves

Figure 3 compares the observed cumulative adoption curves with the predicted curves generated by the Bass Diffusion, Triangle, Circle, and DBM-PD Models.

Comparison of the curves supported the numerical results. The DBM-PD followed the observed cumulative adoption pattern more closely than the other three models across all four datasets. The model was able to capture early growth, mid-stage acceleration, and late-stage saturation with smaller deviation from the observed curve.

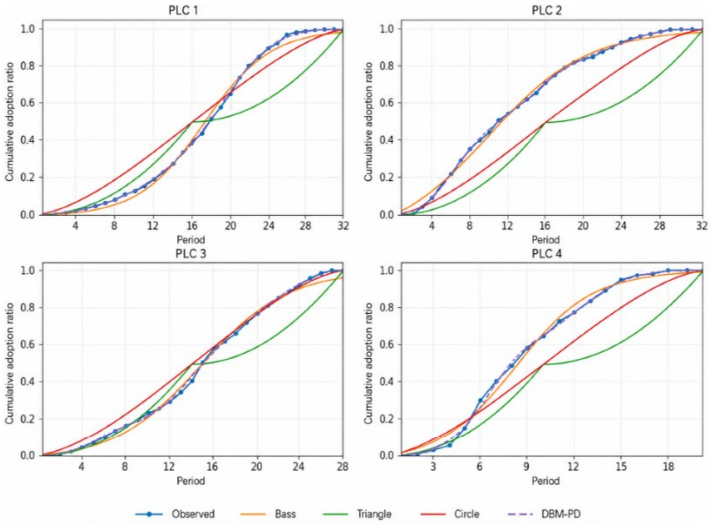


Figure 3. Observed and predicted cumulative adoption curves for PLC 1, PLC 2, PLC 3, and PLC 4

Note: DBM-PD = Diffusion Bubble Model for Product Diffusion; PLC=Product life cycle.

The Bass Diffusion Model also followed the general S-shaped pattern of cumulative adoption. Its performance was stable across the four PLC datasets. However, the Bass diffusion curve was less flexible than the DBM-PD. In some periods, the Bass Diffusion Model slightly underestimated or overestimated the transition of adoption from early growth to maturity. This occurred because the Bass Diffusion Model represents adoption using one continuous behavioral diffusion curve.

The Triangle Model displayed a more rigid pattern. Its prediction curve followed a symmetric structure that did not match the pattern of observed adoption in several datasets. This caused larger deviations, especially when the actual adoption curve increased earlier or later than the assumed midpoint. The Circle Model generated a smoother curve than the Triangle Model, but it still could not fully capture variations in adoption speed across different stages of PLC.

4.5 Discussion of Model Behavior

The findings reported that each model played a different explanatory role in the prediction of PLC. The Bass Diffusion Model provided the strongest behavioral interpretation because p and q represent innovation- and imitation-based adoption within one aggregate S-curve. The DBM-PD did not estimate these behavioral mechanisms. Instead, it represented the observed curve through multiple localized components whose weights captured the relative contribution of adoption at different life-cycle stages. The lower DBM-PD error therefore demonstrated greater representational flexibility, not stronger causal explanation. In this study, the Bass Diffusion Model performed better than the Triangle and Circle Models, suggesting that batik product adoption was more consistent with behavioral diffusion than with simple geometric regularity. This result is consistent with Bass-type PLC modelling [18], although the present study focused on cases in product-level SMEs rather than aggregate market data.

The weaker performance of the Triangle and Circle Models validated that simple geometric assumptions could not fully capture irregular market response in batik fashion products. The Triangle Model assumed a symmetric cumulative adoption pattern, while the Circle Model assumed a smoother bounded accumulation pattern. These assumptions were useful as low-data benchmarks, but they did not explain the causes of adoption. This result is aligned with fashion forecasting studies that described fashion demand as short-lived, uncertain, and sensitive to changing customer preferences. For example, Choi et al. [19] argued that fast fashion forecasting should often be conducted under limited data and designated time, while Yu et al. [20] pointed out that forecasting of fashion sales required efficient methods because conditions of demand and supply were uncertain. Compared with these studies, the present research did not consider advanced AI-based forecasting. Instead, it evaluated whether simple diffusion models could still maintain their effective prediction support for batik fashion SMEs.

The DBM-PD produced the best fitting performance across the four PLC datasets. This result indicated that batik product adoption might contain multiple adoption waves rather than one single diffusion curve. Early adoption may come from loyal customers or direct buyers. Mid-stage adoption may come from broader market exposure, reseller activity, or customer referrals. Late-stage adoption may occur through promotion, repeated interest, or seasonal demand. This staged interpretation is consistent with new product forecasting research that treated demand profiles as patterns that could vary across products. Van Steenbergen and Mes [21] revealed that forecasting of new product demand could benefit from identifying demand profiles based on product characteristics and historical

analogues. The present study differed because the DBM-PD did not depend on large datasets of product attributes. The proposed model built the cumulative adoption curve from weighted diffusion bubbles, thus rendering it more suitable for low-data SME contexts.

The findings also differed from improved Bass studies that relied on external market signals. Fan et al. [22] improved forecasting of product sales by combining the Bass Diffusion/Norton Model with sentiment analysis from online reviews and historical sales data. Similarly, Bruni et al. [9] connected sentiment and uncertainty to the Bass Diffusion Model to improve diffusion forecasting of new energy vehicles. These studies concurred that market signals could strengthen diffusion prediction. However, they required online review data, sentiment processing, or uncertainty modelling. Such requirements may be difficult for batik SMEs as they do not collect structured digital feedback. In contrast, the present study compared models that could be applied with little PLC data and simple parameters, hence introducing a more practical framework for SMEs requiring early-stage prediction under limited-data conditions.

The strong performance of the DBM-PD should be interpreted carefully. The DBM-PD estimated several bubble weights; therefore, it was more flexible than the Bass Diffusion, Triangle, and Circle Models. The sensitivity analysis confirmed that increasing K could further reduce in-sample error, which also maximized the possibility of overfitting. In addition, the empirical evidence was limited to four anonymized products from one batik fashion SME. The cases under study did not capture cross-SME variation in price level, sales channel, intensity of promotion, customer segment, or regional market conditions, and the small sample provided limited power for pairwise statistical tests. As the evaluation adopted complete observed life cycles; it could not demonstrate out-of-sample forecasting from genuinely partial early-stage data. The DBM-PD should therefore be viewed as the best in-sample fitting model for these four cases, rather than a universally validated forecasting model. Broader multi-SME datasets, rolling-origin validation, and training windows based on 30%, 50%, and 70% of each life cycle were necessitated before claiming general applicability.

4.6 Practical Implications for Batik Fashion Small and Medium Enterprises

The results provided practical implications for batik fashion SMEs.

First, the models could help SMEs estimate the PLC of new batik products under limited-data conditions. This is important because SMEs often need to decide production volume, stock allocation, and timing of product renewal prior to the availability of complete sales data.

Second, the Bass Diffusion Model could be used when SMEs need a simple behavioral diffusion model. Its parameters provide useful interpretation. A higher innovation effect may indicate stronger early product appeal, while a higher imitation effect may indicate stronger market response through customer interaction or word-of-mouth.

Third, the Triangle and Circle Models could be implemented as simple reference models when SMEs possessed very limited sales-related information. These models could perform calculation easily and provide an initial expectation of cumulative adoption. However, they should be used with caution because their assumptions may not match real-market behavior.

Fourth, the DBM-PD could support more flexible prediction when SMEs carry out early sales observations or have comparable patterns of PLC. The model could capture multiple adoption phases and assist in identifying whether demand is concentrated in the early stage, middle stage, or late stage of the PLC. An illustrative example for its implementation is as follows:

Assume that an SME expects a new batik product to have a market potential of $M = 500$ units over a 32-period life cycle. After period 8, cumulative sales are 100 units, so the observed cumulative adoption ratio is 0.2. The SME enters the eight period-level observations into the DBM-PD routine, retains the fixed seven-bubble configuration, estimates the non-negative weights, and updates the cumulative forecast. Suppose the illustrative output gives $\hat{R}(16) = 0.55$ and $\hat{R}(24) = 0.82$. The expected additional demand is therefore $(0.55 - 0.20) \times 500 = 175$ units before period 16 and $(0.82 - 0.55) \times 500 = 135$ units from period 16 to 24. If usable inventory plus work in process is only 120 units, the SME identifies an initial shortfall of 55 units and can release a controlled additional batch rather than producing the entire remaining demand at once. The weights and demand increments can then be re-estimated after each new sales period. These numbers are illustrative but the procedure exhibits how the cumulative curve could be translated into rolling production, inventory, and decisions of product renewal.

4.7 Theoretical and Methodological Implications

The findings contribute to the prediction of technology-based PLC by distinguishing behavioral interpretation from the representation of flexible curves. The Bass Diffusion Model explains adoption through innovation and imitation parameters; the Triangle and Circle Models impose transparent geometric shapes and the DBM-PD decomposes adoption into weighted stage-specific components. This distinction helps decision-makers select a model according to their priority of behavioral explanation, minimal data requirements, or close representation of irregular staged adoption.

This comparison established a complete methodological framework instead of using one model alone. If only the Bass Diffusion Model is employed, the analysis may ignore simple geometric benchmarks. If geometric models are used, the analysis may ignore behavioral diffusion effects. If only the DBM-PD is used, the analysis may produce strong fitting results without conducting enough theoretical comparison. By comparing all four models, this study carried out a more balanced evaluation of accuracy, interpretability, and practical usability.

The results implied that model selection should not depend merely on error values. For SMEs, the best model should be comprehensible and usable. The DBM-PD produced the lowest error but the Bass Diffusion Model offered stronger behavioral interpretation. The Triangle and Circle Models were less accurate but they remained useful as simple low-data benchmarks. Therefore, the choice of model should depend on the context of decision, availability of data, and required level of interpretation.

5 Conclusions

This study compared four diffusion models for low-data PLC prediction in batik fashion products. The models included the Bass Diffusion model, Triangle model, Circle model, and DBM-PD. Four datasets of PLC from batik fashion products were used as empirical cases. Each dataset was transformed into a cumulative adoption ratio so that all models could be evaluated on the same scale.

The results recorded that the DBM-PD produced the best in-sample fitting performance across all four PLC datasets. Its average RMSE was 0.0087, compared with 0.0254 for Bass, 0.0964 for Circle, and 0.1635 for Triangle. The identical model ranking across all four cases was supported by the Friedman test, $\chi^2(3) = 12.00$, $p = 0.007$, with Kendall's $W = 1.00$. The sensitivity analysis further confirmed that five bubbles underfit the staged adoption patterns, whereas nine narrow bubbles could reduce in-sample error at the expense of additional parameters. The seven-bubble, $\sigma = 0.08$ configuration was therefore retained as a parsimonious specification.

The Bass Diffusion Model kept its role as the strongest behavioral benchmark. Its better performance compared with the Triangle and Circle Models illustrated that batik product adoption was influenced by innovation and imitation effects. This meant that market response, customer interaction, and word-of-mouth played a crucial role in the new batik fashion product diffusion. In contrast, the Triangle and Circle Models were effective as simple low-data benchmarks, but their geometric assumptions were less able to capture irregular adoption patterns.

The key contribution of this study is a transparent comparison of behavioral, geometric, and decompositionbased diffusion assumptions for batik fashion SMEs. The DBM-PD presents the closest curve representation and a practical basis for rolling production and inventory updates, while the Bass Diffusion Model remains more interpretable in terms of innovation and imitation. The proposed framework could aid the planning of production, stock control, launch timing, and product-renewal decisions, but the empirical evidence remains case-based.

The conclusions are confined by only four anonymized product cases from one SME, the inaccessibility of product-level commercial attributes, and the request for complete life-cycle data for in-sample evaluation. Future research should test more products across multiple SMEs and channels, use rolling-origin or holdout validation, and estimate the DBM-PD from partial training windows such as 30%, 50%, and 70% of the life cycle. External market signals may also be integrated, provided that the resulting model remains usable for SMEs with limited data and computational resources.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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